

Tomás David Navarro Múnera

DEVOPS ENGINEER · SECURITY RESEARCHER

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Summary

DevOps Engineer and Security Researcher who graduated with an M.A.Sc. in Information System Security from Concordia University, with a research focus on container security. Experienced in automating and securing AWS infrastructure with Terraform, Ansible, Docker, and Kubernetes, and in researching kernel and cross-container attacks as a Mitacs-funded Graduate Research Student with the NSERC/Ericsson IRC in SDN/NFV Security. Computer Science graduate from EAFIT University with a growing focus on offensive security, gained through CTF competitions and hands-on penetration testing.

Education

Concordia University

M.A.Sc. IN INFORMATION SYSTEM SECURITY

Montreal, Quebec, Canada

May 2023 - Aug. 2026

EAFIT University

B.S. IN COMPUTER SCIENCE AND ENGINEERING, GPA: 4.28/5

Medellín, Colombia

Jan. 2018 - Dec. 2022

Experience

Concordia University

GRADUATE RESEARCH STUDENT - NSERC/ERICSSON IRC IN SDN/NFV SECURITY

Montreal, Quebec, Canada · Hybrid

May. 2023 - Aug. 2026

- Selected for a Mitacs Accelerate research internship (Sep. - Dec. 2023) with Ericsson.
- Use of Docker and Podman for testing cross-container vulnerabilities and exploits.
- Use of eBPF for detecting kernel attacks and mitigation solutions.
- Research Kernel and cross-container attacks.

Pyxis SAS

DEVOPS ENGINEER

Medellín, Colombia

May. 2022 - March. 2023

- Use of Terraform and Terragrunt for IaC construction and deployment in AWS.
- Use of Ansible as an automation tool for deployment of server tools and configuration.
- AWS Networking and Configuration for High Availability and High Scalability in applications.
- Use of Docker for environmental configuration and deployment of applications in AWS.

Empresas Públicas de Medellín

DATA SCIENTIST INTERN

Medellín, Colombia

Aug. 2021 - Feb 2022

- Use of Data Science programming languages [Python, Matlab, R].
- Build Machine Learning Models for hydrological forecasting.
- Data Analysis of the models' results.
- Build WebScraping Tool to extract data from XM.com.co.
- Build API connector tool to get data from XM.com.co to analyse data.

EAFIT University, Apolo Supercomputing Center

RESEARCH ASSISTANT

Medellín, Colombia

Sep. 2019 - Jun. 2021

- Linux System Admin, review packages for vulnerabilities.
- Deployment of private web-apps in Docker to secure the cluster.
- Created "LMOD module generator" script for apps usage [Bash].
- Use of Ansible and Vagrant as an automation tool for deployment of server tools and configuration.
- Install HPC applications for cluster users.

Publications

KubeBPF: eBPF-Based In-Kernel Machine Learning Approach for Detecting and Mitigating Abstract Resource Attacks for Containers

M.A.Sc. THESIS · CONCORDIA UNIVERSITY

Aug. 2026

Accepted

CONnecting The EXtra doTS (CONTEXTS): Correlating External Information about Points of Interest for Attack Investigation

SAREH MOHAMMADI, HUGO KERMABON-BOBINNEC, AZADEH TABIBAN, LINGYU WANG, TOMÁS NAVARRO MÚNERA, YOSR JARRAYA

IEEE S&P 2025

Accepted

Skills

DevOps	AWS, Docker, Vagrant, Linux, eBPF, Ansible, Kubernetes, Terraform, Terragrunt
Back-end	Laravel, Django, REST API
Front-end	Bootstrap, HTML5
Programming	Python, JAVA, LaTeX, C++, C, PHP, R, Matlab, GTK3
Languages	Spanish (Native), English (Fluent), French (Basic)

Extracurricular Activity

Various CTFs

PARTICIPANT

- Participated in NorthSec 2024 and NorthSec 2025 CTFs.
- Participated in PolyPwn 2025 and 2026 CTF (Third place in the 2026 edition).
- Participated in @Hack 2024 CTF.

Montreal, Canada

Feb. 2024 - Present

AtHack CTF

ATHACK ORGANIZER

- Helped organize @Hack CTF event where more than 300 participated.
- Gave support on challenges to participants.

Montreal, Canada

Feb. 2025

Courses & Certificates

SOC Level 1 Learning Path

CERTIFICATE

TryHackMe

Sep. 2026

Cloud and DevOps Fundamentals. Cloud and Automation tools.

CERTIFICATE

EPAM Learning

Apr. 2025

AWS Certified Cloud Practitioner

CERTIFICATE

Amazon Web Services

May. 2023

Complete Ethical Hacking Bootcamp 2022: Zero to Mastery

CERTIFICATE

Udemy

Jun. 2022

DevOps in the Cloud with Terraform, Ansible, and Jenkins

CERTIFICATE

Udemy

Jun. 2022

- Deployed Grafana and Prometheus to EC2 instances using DevOps tools Terraform, Ansible, and Jenkins.